

USB Charging

All About USB Charging



REVIEW / UPM USB METER OR TRIGGER

KEWEISI KWS-1902C Type-C USB Tester UPM Review

🐦 DECEMBER 14, 2019 OCTOBER 17, 2022 🐦 USBCHARGING 🐦 16 COMMENTS

TL;DR: ROAAAR! kws-1802cmon shinka! KWS-1902CMON!! However, the 65.536W zeroing bug still exists.

Official specifications

- Product Model: KWS-1902C
- Screen Type: 0.96 inches color screen
- Voltage measurement range: 4-30V
- Voltage resolution: 0.002V
- Current measurement range: 0-5.1A
- Current resolution: 0.002A
- Power measurement range: 0-155W
- Energy accumulation range: 0-999999mWh
- Load resistance: 1-9999.90Ω
- Temperature range: 0°C ~ 80°C/32°F ~ 176°F
- Timing: 0-99 hours
- Interface: USB-C 24pin
- Testing direction: bidirectional
- Color: Black, White
- Package weight: 15g
- Product size: 55mm × 26.5mm × 11.5mm





Click on image to enlarge.



User Interfaces

Main Display

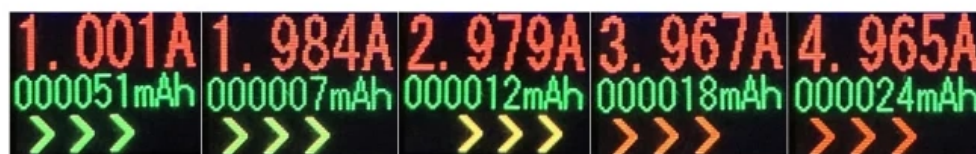
MAIN DISPLAY



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Current Direction Indicator

Current Direction Indicator



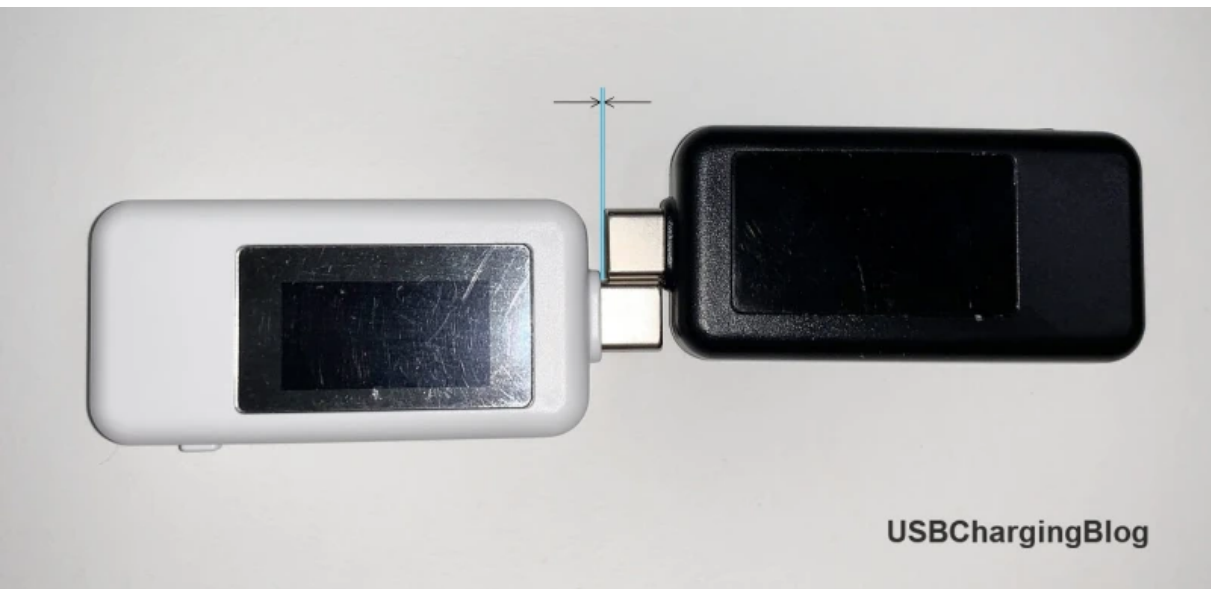
Color: Green → Orange

Speed: Normol → Fast

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The larger the current, the closer the color is to orange and the faster the moving speed.

KWS-1902C VS KWS-1802C



They are almost the same size, except that the Type-C plug of 1902C is slightly shorter.



What's New

KWS-1902C is the upgrade to KWS-1802C, and it has

- Color LCD screen
- Higher accuracy and resolution
- Energy measurement
- D+ and D- voltages measurement
- Load resistance measurement
- Voltage calibration

What's Removed

- Data group

Pinouts

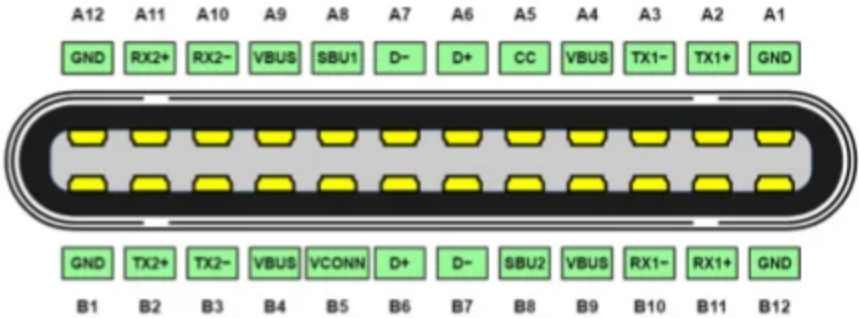


It comes with full-featured USB Type-C connectors.

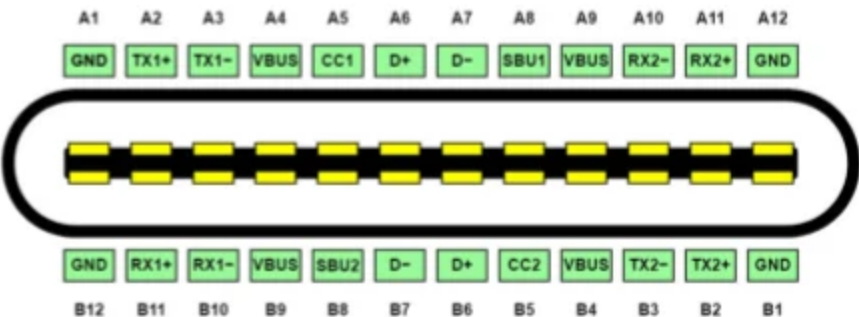
Wired



Plug



Receptacle



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Click on image to enlarge.

Measurements

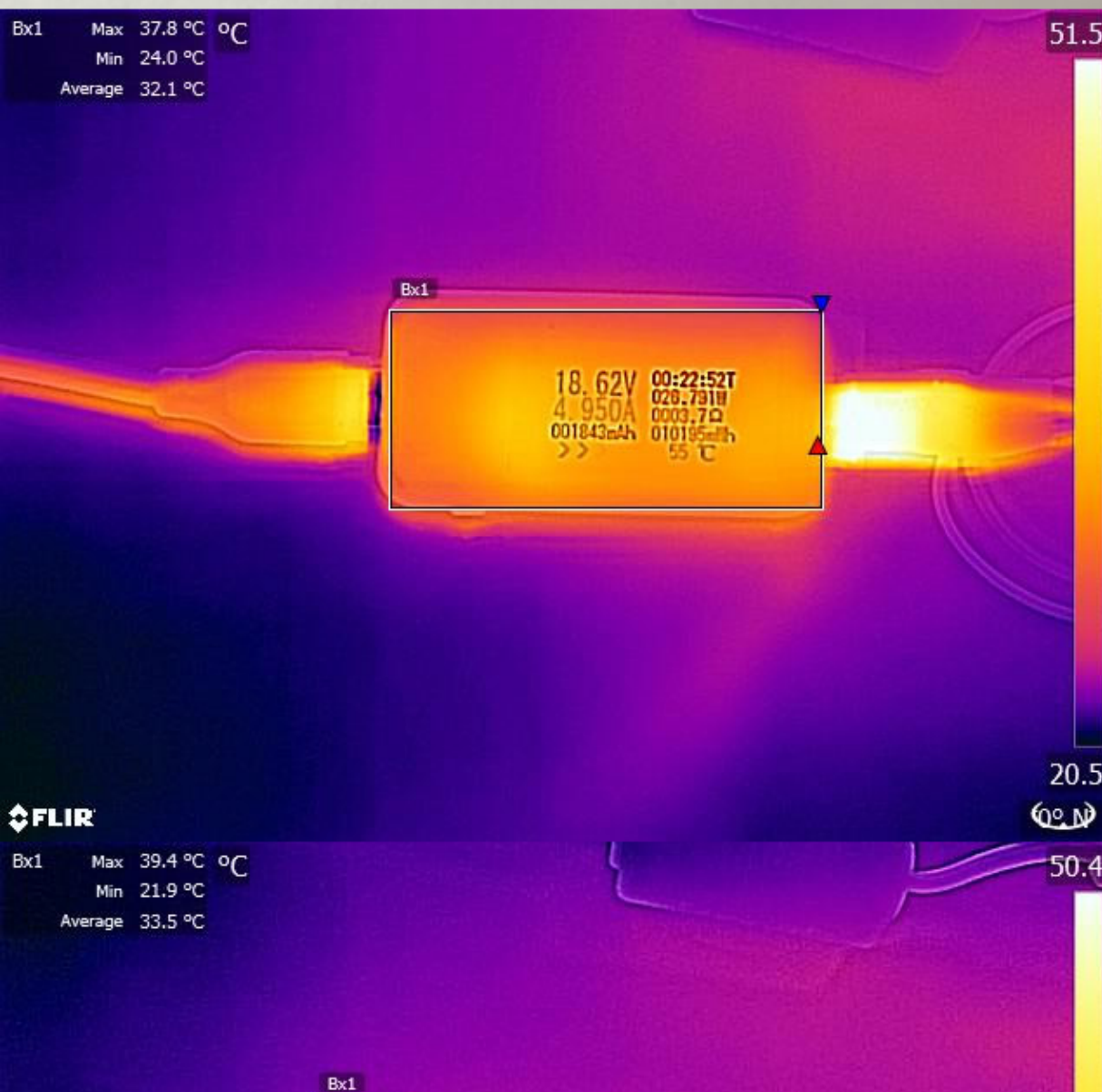
YMMV as always.

Voltage Accuracy (Unloaded, Ambient Temp 17.7°C/63.86°F)	
V	V _{1902C}
2.0001	—
2.8001	2.771
3.0001	2.971

4.0001		3.965
5.0002		4.956
9.0002		8.938
10.0002		9.940
12.0001		11.93
15.0003		14.91
18.0000		17.89
20.0002		19.88
Voltage Accuracy (Calibrated, Unloaded, Ambient Temp 17.7°C/63.86°F)		
V		V1902C
2.0000		–
2.8002		2.797
3.0003		2.997
4.0001		4.000
5.0002		5.000
9.0003		9.016
10.0001		10.02
12.0002		12.03
15.0001		15.04
18.0003		18.05
20.0001		20.05
Current Accuracy (Ambient Temp 18.2°C/64.76°F)		
V	A	A1902C
5	0.00501	0.000
	0.01000	0.000
	0.05001	0.048
	0.10002	0.097
	0.50003	0.495
	1.00001	0.992
	2.00001	1.985
	3.00004	2.984
	4.00003	3.979
	5.00002	4.970
Own Current Consumption		
V	mA	
4-20	~16.9	

- The usb-c connector is reversible; the arrow indicates the direction of the current.
- The plug is for output and the receptacle is for input.
- Screen orientation, charging time (T), capacity (mAh), and energy (mWh) will be saved after power off.
- The lowest current it can detect is around 0.014A.
- The accuracy of current is acceptable.
- It takes about 2.8V to work completely.
- The voltage is not very accurate, but it gets better after calibration.
- Own current consumption is near 16.9mA.
- $R_{VBUS} = 38.7\text{m}\Omega$, $R_{GND} = 41.5\text{m}\Omega$, so the internal resistance = $80.2\text{m}\Omega$ (Includes both connectors).
- The time begins to count only when a current is detected.

Temperature





18.6V 5A @18.2°C/64.76°F over 20 min

The temperature of the current shunt resistor is within 65°C/149°F when the reading on the display is 55°C.

Tear-Down



MCU: Nuvoton N76E003AT20

Bugs

65.536W Zeroing Bug



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The 65.536W Zeroing Bug is a legacy bug. It was found in some 1802Cs with a specific font. You do the math when the power exceeds 65.535W. Correct reading = current reading + 65.536W. Looks like all the 1902Cs use the same font. That means this bug exists in all 1902Cs. Check yours! Lol.

Conclusion

The 1902C is the upgrade to 1802C. It comes with color display, higher accuracy, and higher resolution. The mWh measurement, which I mentioned in the 1802C review, finally comes out. It's OK for measurements below 65W. Because of the 65.536W Zeroing Bug, you need to do some math when the power goes beyond 65.535W. Right, DO! THE! MATH!

Related Review

[KEWEISI KWS-1802C Type-C USB Tester UPM First Look](https://usbchargingblog.wordpress.com/2018/10/15/keweisi-kws-1802c-type-c-usb-tester-upm-first-look/)

(<https://usbchargingblog.wordpress.com/2018/10/15/keweisi-kws-1802c-type-c-usb-tester-upm-first-look/>)

➤ [1902C](#), [KEWEISI](#), [KWS-1902C](#), [UPM](#), [USB METER](#), [USB TESTER](#)



16 thoughts on “KEWEISI KWS-1902C Type-C USB Tester UPM Review”

TNT says:

APRIL 7, 2023 AT 9:34 AM

Have you tried “KWS 066C” yet? it has QC/PD/VOOC support.

REPLY 🐦

UsbCharging says:

APRIL 19, 2023 AT 7:46 PM

The 65.536W Zeroing Bug seems to be fixed on KWS 066C.

https://usbchargingblog.files.wordpress.com/2023/04/kws-066c_ed.jpg

REPLY 🐦

Amir says:

NOVEMBER 7, 2021 AT 3:45 PM

Support the power delivery?

REPLY 🐦

UsbCharging says:

NOVEMBER 7, 2021 AT 3:48 PM

Yes, but it can only indicate V, A, and W. It is up to the user to determine whether PD happens. No PD trigger.

REPLY 🐦

3. **USB charging in 2021 | Nelson's log** says:

OCTOBER 9, 2021 AT 12:46 PM

[...] charger Amazon sent me is called a KWS-1902C, there's a nice review here. It claims to support up to 150W, 4-30V, and 0-5A. The first look on the blog is from October 2018, [...]

REPLY 🐦

何先生 says:

JULY 27, 2021 AT 1:02 AM

顯示電壓比實際輸入的還低 該如何校正?

REPLY 🐦

UsbCharging says:

JULY 27, 2021 AT 4:47 PM

You will need a power source and a multimeter to do this. If possible, make sure that your ambient temperature is approximately 23 degrees Celsius.

1. Connect the power source, KWS-1902C USB tester and multimeter as shown in the figure.

https://usbchargingblog.files.wordpress.com/2021/07/how_to_cal_kws-1902c_01.png

2. Adjust the power source until the multimeter reads 5V as shown. (Because my KWS-1902C has been calibrated before, its reading is consistent with the reading of my multimeter.) 🐦

https://usbchargingblog.files.wordpress.com/2021/07/how_to_cal_kws-1902c_02.jpg

3. Unplug the KWS-1902C from the power source.

4. Plug it back to the the power source with pressing and holding the button on the KWS-1902C.

Translation:

您將需要一個電源和一個萬用表來執行此操作。 如果可能，請確保您的環境溫度約為 23 攝氏度。

1. 如圖所示連接電源、KWS-1902C USB 測試儀和萬用表。

https://usbchargingblog.files.wordpress.com/2021/07/how_to_cal_kws-1902c_01.png

2. 調整電源直到萬用表讀數為 5V，如圖所示。（因為我的KWS-1902C之前已經校準過，所以它的讀數和我萬用表的讀數是一致的。） https://usbchargingblog.files.wordpress.com/2021/07/how_to_cal_kws-1902c_02.jpg

3. 拔掉 KWS-1902C 的電源。

4. 按住 KWS-1902C 上的按鈕，將其插回電源。

REPLY 🐦

Paul says:

JUNE 12, 2021 AT 3:32 AM

What is the best usc-c tester?

The description on Amazon it can display the charging protocol, but not sure how that is done.

REPLY 🐦

UsbCharging says:

JUNE 12, 2021 AT 10:49 AM

Hello! There's no best USB-C tester currently. It depends on what you mainly use it for and what features are most important to you.

Nope, KWS-1902C cannot display the charging protocol.

REPLY 🐦

Tiago says:

OCTOBER 19, 2022 AT 5:09 PM

Mine in the beginning was showing correct values but now most of the times it shows wrong values... Even making factory reset don't solve. There's something I'm missing or something more I can do?

UsbCharging says:

OCTOBER 19, 2022 AT 5:20 PM

Does calibration help?

Tiago says:

OCTOBER 19, 2022 AT 5:30 PM

For calibration you mean factory settings?

UsbCharging says:

OCTOBER 19, 2022 AT 11:02 PM

You will need a power source and a multimeter to do this. If possible, make sure that your ambient temperature is approximately 23 degrees Celsius.

1. Connect the power source, KWS-1902C USB tester and multimeter as shown in the figure.

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https://usbchargingblog.files.wordpress.com/2021/07/how_to_cal_kws-1902c_02.jpg 🐦

3. Unplug the KWS-1902C from the power source.

4. Plug it back to the the power source with pressing and holding the button on the KWS-1902C.

eddy says:

JANUARY 26, 2021 AT 11:13 PM

Hei, does it work in pd protocol?

REPLY 🐦

UsbCharging says:

JANUARY 27, 2021 AT 12:18 PM

Yes, but it can only indicate V, A, and W. It is up to the user to determine whether PD has occurred.

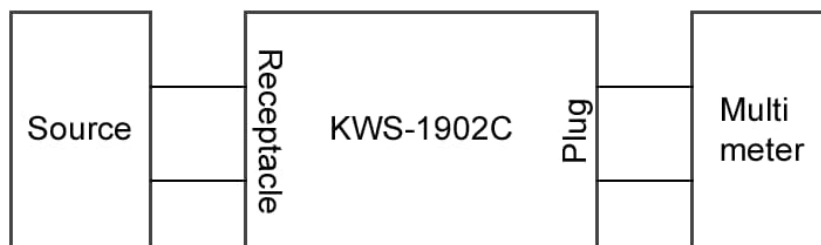
REPLY 🐦

eddy says:

FEBRUARY 4, 2021 AT 11:58 PM

Thanks. I'll buy it anyway, because it's the cheapest type c meter in my country market

[START A BLOG AT WORDPRESS.COM.](https://www.wordpress.com)



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